

Gaussian stress-test methodology

Goal: How well does the inverse + surrogate pipeline generalize to Quantum Metal parameters it has never seen before

Probe the in-between regions of SQuADDS by perturbing held-out test points with controllable Gaussian noise.

1. Pick seeds

Choose 30 held-out test samples from SQuADDS and treat each one as the center of a Gaussian ball in Quantum Metal parameter space.

2. Add Gaussian noise

For each parameter i :

$$p_i^{\text{new}} = p_i^{\text{orig}} + \mathcal{N}(0, \sigma_i)$$

with $\sigma_i = f \cdot \text{range}(p_i)$.

Reject samples that leave the box or violate layout constraints.

3. Sweep sigma

Sweep f across

$\{1\%, 2\%, 5\%, 10\%, 20\%\}$
of each parameter's range.

Draw 6,000 samples per σ
→ 30,000 total.

Surrogate evaluates all 30,000 samples in seconds (~ 5000 Ansys-hours equivalent).